

Nikhil Thakur

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EDUCATION

Indian Institute of Engineering Science and Technology

B.Tech. Information Technology

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Current CGPA: 8.45/10.0

TECHNICAL SKILLS

Languages: C, C++, JavaScript, Typescript, python

Backend: Node.js, Express.js

Frontend: React, TailwindCSS, HTML, CSS, Bootstrap

Databases: MongoDB

Developer Tools: Postman, VS Code, GitHub

PROJECTS

Online Judge Platform | *MERN Stack, Docker, AWS, JWT, Nginx, Gemini API*

GitHub

completed

- Engineered a full-stack coding platform enabling users to write, execute, and submit solutions with automated evaluation.
- Designed a clean and interactive interface using React and Tailwind CSS for problem browsing and code submission workflows.
- Built backend services with Node.js and Express to manage authentication, problem lifecycle, and submission processing.
- Developed an isolated execution environment supporting multiple languages with detailed handling of compilation and runtime failures.
- Structured data models in MongoDB for efficient storage of users, problems, submissions, and test cases.
- Enabled administrative controls for managing problem statements and test cases via JSON import/export.
- Integrated AI-driven assistance for explanation, debugging, and optimization of user code using Gemini API.
- Deployed the system on AWS with containerization and reverse proxy setup for secure and scalable access.

Expense Tracker Application | *MERN Stack, TypeScript, JWT, Google Analytics*

GitHub

completed

- Developed a full-stack expense tracking application with a responsive UI using React and Tailwind CSS.
- Implemented JWT-based authentication and built RESTful APIs for transaction management with filtering and pagination.
- Designed MongoDB schemas for efficient storage and querying of user and transaction data.
- Integrated Google Analytics to track real-time user activity and user interaction events such as login and transactions.

Real-Time Chat Application | *MERN Stack, Socket.IO, Redis, JWT*

GitHub

completed

- Engineered a real-time messaging system enabling seamless bidirectional communication between users.
- Built a responsive client interface with React and Tailwind CSS, ensuring smooth navigation across devices.
- Implemented event-driven communication using Socket.IO with features like typing indicators and user presence tracking.
- Developed RESTful services with Node.js and Express to manage users, conversations, and message flow.
- Modeled and optimized data storage in MongoDB for efficient querying of chats and message history.
- Leveraged Redis to manage active sessions and support real-time event handling.
- Secured the system using token-based authentication with JWT.
- Ensured reliable delivery by handling message persistence and synchronization for offline users.

ACHIEVEMENTS

- Codeforces Pupil** (max rating **1200+**) and achieved **1700+ rating** on **LeetCode**.
- Solved **900+ problems** across platforms including **LeetCode**, **Codeforces**, and **GeeksforGeeks**.
- Active member of **Google Developer Group (GDG)**, **IIST Shibpur**, participating in technical events and workshops.